Income and Substitution Effects in a Two-Period Consumption Model

1 The Consumer's Maximization Problem

Consider a consumer who lives for two periods, indexed by $t = 0, 1$. The consumer receives an exogenous income stream (y_0, y_1) and chooses consumption (c_0, c_1) to solve

$$\max_{c_0, c_1} U(c_0, c_1)$$

subject to the intertemporal budget constraint

$$c_0 + \frac{c_1}{R} = W, \qquad \boxed{W = y_0 + \frac{y_1}{R}},$$

where W is the *present value of the consumer's income stream* and $R = 1 + r > 0$ is the gross real interest rate.

The Lagrangian is

$$\boxed{\mathcal{L}(c_0, c_1, \lambda) = U(c_0, c_1) + \lambda \left(W - c_0 - \frac{c_1}{R} \right).}$$

Assume initially that preferences are strictly increasing, strictly quasi-concave, and differentiable, and that the optimum is interior. The first-order conditions are

$$U_0(c_0, c_1) = \lambda, \qquad U_1(c_0, c_1) = \frac{\lambda}{R}, \qquad c_0 + \frac{c_1}{R} = W.$$

Thus,

$$\boxed{\frac{U_0(c_0, c_1)}{U_1(c_0, c_1)} = R.}$$

The marginal rate of substitution between current and future consumption equals the gross real interest rate.

2 Marshallian Demands and Indirect Utility

The solution depends on wealth W and the interest rate R. It gives the *Marshallian demand functions*:

$$\boxed{c_0 = c_0(W, R), \qquad c_1 = c_1(W, R).}$$

Substituting these demands into utility gives the *indirect utility function*:

$$\boxed{V(W,R) = U(c_0(W,R), c_1(W,R)).}$$

The indirect utility function gives the maximum utility attainable at a given wealth and interest rate. The direct utility function describes preferences over consumption bundles; the indirect utility function describes welfare as a function of the economic opportunities available.

Roy's Identity

More generally, let $\mathbf{p} = (p_0, p_1)$ denote the price vector, and let $V(\mathbf{p}, W)$ be indirect utility. *Roy's identity* recovers Marshallian demands:

$$\boxed{c_i(\mathbf{p}, W) = -\frac{\partial V/\partial p_i}{\partial V/\partial W}.}$$

By the envelope theorem, the optimal Lagrange multiplier satisfies

$$\boxed{\lambda^* = \frac{\partial V(W,R)}{\partial W}.}$$

Thus, λ^* is the *marginal utility of wealth*: the increase in maximized utility resulting from an additional unit of present-value wealth.

3 The Expenditure Function

Fix a utility level $\bar{u}$. Instead of asking how much utility the consumer can attain with given wealth, ask: *What is the minimum wealth necessary to attain $\bar{u}$?*

The *expenditure function* is

$$\boxed{E(\bar{u}, R) = \min_W \{W : V(W,R) \geq \bar{u}\}.}$$

It gives the minimum present-value wealth required to attain $\bar{u}$ at interest rate R. Since indirect utility is strictly increasing in wealth,

$$\boxed{V(E(\bar{u}, R), R) = \bar{u}.}$$

Thus, indirect utility and expenditure are inverses with respect to wealth.

4 Hicksian Demands and Shephard's Lemma

The *Hicksian demands* $h_0(\bar{u}, R)$ and $h_1(\bar{u}, R)$ minimize expenditure while attaining $\bar{u}$. Minimum expenditure satisfies

$$E(\bar{u}, R) = h_0(\bar{u}, R) + \frac{h_1(\bar{u}, R)}{R}.$$

More generally, with price vector $\mathbf{p}$, *Shephard's lemma* states

$$\boxed{h_i(\mathbf{p}, \bar{u}) = \frac{\partial E(\mathbf{p}, \bar{u})}{\partial p_i}.}$$

Roy's identity recovers Marshallian demands from indirect utility; Shephard's lemma recovers Hicksian demands from the expenditure function. Marshallian demand holds wealth fixed in a price comparison, while Hicksian demand holds utility fixed.

5 Derivation of the Slutsky Equation

The two demand systems satisfy

$$\boxed{h_i(\mathbf{p}, \bar{u}) = c_i(\mathbf{p}, E(\mathbf{p}, \bar{u})).}$$

Differentiate with respect to p_j, holding $\bar{u}$ fixed:

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial c_i}{\partial p_j} + \frac{\partial c_i}{\partial W}\frac{\partial E}{\partial p_j}.$$

By Shephard's lemma, $\partial E/\partial p_j = h_j$. At the initial optimum, $h_j = c_j$. Rearranging gives the *Slutsky equation*:

$$\boxed{\underbrace{\frac{\partial c_i}{\partial p_j}}_{\text{Total price effect}} = \underbrace{\frac{\partial h_i}{\partial p_j}}_{\text{Substitution effect}} - \underbrace{c_j \frac{\partial c_i}{\partial W}}_{\text{Income-effect term}}.}$$

The substitution effect holds utility constant; the income effect captures the change in purchasing power from a price change.

6 Three Examples

We solve for Marshallian and Hicksian demands under three preference specifications. Assume $W > 0$, $R > 0$, and nonnegative consumption.

6.1 Perfect Substitutes

Suppose

$$U(c_0, c_1) = c_0 + c_1.$$

The consumer maximizes utility subject to $c_0 + c_1/R = W$. When $R > 1$, future consumption is cheaper in present-value terms; when $R < 1$, current consumption is cheaper. Marshallian demands are

$$\boxed{(c_0^M, c_1^M) = \begin{cases} (W, 0), & R < 1, \\ (0, RW), & R > 1. \end{cases}}$$

At $R = 1$, every nonnegative bundle with $c_0 + c_1 = W$ is optimal.

Hicksian demand minimizes $c_0 + c_1/R$ subject to $c_0 + c_1 \geq \bar{u}$:

$$\boxed{(h_0, h_1) = \begin{cases} (\bar{u}, 0), & R < 1, \\ (0, \bar{u}), & R > 1. \end{cases}}$$

At $R = 1$, any nonnegative bundle with $c_0 + c_1 = \bar{u}$ minimizes expenditure. Hence

$$\boxed{E(\bar{u}, R) = \bar{u} \min\left\{1, \frac{1}{R}\right\}.}$$

Perfect substitutes generally produce corner solutions.

6.2 Log-Linear Preferences

Suppose

$$\boxed{U(c_0, c_1) = \ln c_0 + \beta \ln c_1, \qquad 0 \leq \beta \leq 1.}$$

For $\beta > 0$, the first-order conditions are

$$\frac{1}{c_0} = \lambda, \qquad \frac{\beta}{c_1} = \frac{\lambda}{R},$$

so $c_1 = \beta R c_0$. Substituting into the budget constraint yields

$$\boxed{c_0^M(W, R) = \frac{W}{1+\beta}, \qquad c_1^M(W, R) = \frac{\beta R W}{1+\beta}.}$$

For Hicksian demand, minimize expenditure subject to

$$\ln c_0 + \beta \ln c_1 \geq \bar{u}.$$

The tangency condition again gives $c_1 = \beta R c_0$. At the optimum,

$$\ln c_0 + \beta \ln(\beta R c_0) = \bar{u},$$

so

$$\boxed{h_0(\bar{u}, R) = \frac{e^{\bar{u}/(1+\beta)}}{(\beta R)^{\beta/(1+\beta)}}}, \qquad \boxed{h_1(\bar{u}, R) = (\beta R)^{1/(1+\beta)} e^{\bar{u}/(1+\beta)}}.$$

The expenditure function is

$$\boxed{E(\bar{u}, R) = (1+\beta) \frac{e^{\bar{u}/(1+\beta)}}{(\beta R)^{\beta/(1+\beta)}}.}$$

At $\beta = 0$, future consumption has no utility value: $c_0^M = W$, $c_1^M = 0$, $h_0 = e^{\bar{u}}$, and $h_1 = 0$. The formulas above for $\beta > 0$ have these continuous limits.

6.3 Perfect Complements

Suppose

$$U(c_0, c_1) = \min\{c_0, c_1\}.$$

The consumer wants $c_0 = c_1 = c$. Since $c + c/R = W$,

$$\boxed{c_0^M(W, R) = c_1^M(W, R) = \frac{RW}{R+1}.}$$

To attain $\bar{u}$, the consumer must have $c_0 \geq \bar{u}$ and $c_1 \geq \bar{u}$. Thus,

$$\boxed{h_0(\bar{u}, R) = h_1(\bar{u}, R) = \bar{u}}, \qquad \boxed{E(\bar{u}, R) = \bar{u}\left(1 + \frac{1}{R}\right).}$$

Perfect complements have zero substitution effects because their optimal consumption proportions do not change.

7 The Slutsky Equation in an Endowment Economy

Let the price of future consumption be $p_1 = 1/R$. In an endowment economy, wealth depends on that price:

$$W = y_0 + p_1 y_1.$$

Applying the chain rule to Marshallian demand along the endowment-induced wealth change,

$$\frac{dc_0}{dp_1} = \frac{\partial c_0}{\partial p_1} + y_1 \frac{\partial c_0}{\partial W}.$$

Substituting the ordinary Slutsky equation,

$$\frac{\partial c_0}{\partial p_1} = \frac{\partial h_0}{\partial p_1} - c_1 \frac{\partial c_0}{\partial W},$$

gives the *endowment-economy Slutsky equation*:

$$\boxed{\underbrace{\frac{dc_0}{dp_1}}_{\text{Total effect}} = \underbrace{\frac{\partial h_0}{\partial p_1}}_{\text{Substitution effect}} + \underbrace{(y_1 - c_1)\frac{\partial c_0}{\partial W}}_{\text{Net income effect}}.}$$

The net income effect depends on the consumer's endowment of future consumption relative to optimal future consumption.

8 A Numerical Example

Consider log-linear preferences with

$$\boxed{\beta = 0.97, \qquad R = 1.05, \qquad y_0 = 4, \qquad y_1 = 6.}$$

Present-value wealth is

$$W = 4 + \frac{6}{1.05} = \frac{68}{7} \approx 9.7143.$$

The Marshallian demands are

$$\boxed{c_0 = \frac{W}{1+\beta} \approx 4.9311, \qquad c_1 = \frac{\beta R W}{1+\beta} \approx 5.0223.}$$

Since $c_0 > y_0 = 4$, the consumer is initially a net borrower.

The price of future consumption is

$$p_1 = \frac{1}{1.05} \approx 0.95238.$$

For log-linear preferences, Hicksian demand for current consumption is

$$h_0(\bar{u}, p_1) = e^{\bar{u}/(1+\beta)} \left(\frac{p_1}{\beta}\right)^{\beta/(1+\beta)}.$$

Differentiating at fixed utility gives

$$\frac{\partial h_0}{\partial p_1} = \frac{\beta}{1+\beta} \frac{h_0}{p_1}.$$

At the initial optimum $h_0 = c_0$, so the substitution effect is

$$\boxed{\frac{\partial h_0}{\partial p_1} \approx 2.5494.}$$

The net income effect is

$$\boxed{(y_1 - c_1)\frac{\partial c_0}{\partial W} = \frac{6 - 5.0223}{1.97} \approx 0.4963.}$$

Consequently,

$$\boxed{\underbrace{3.0457}_{\text{Total price effect}} = \underbrace{2.5494}_{\text{Substitution effect}} + \underbrace{0.4963}_{\text{Net income effect}}.}$$

Because $p_1 = 1/R$, $dp_1/dR = -1/R^2$, so

$$\boxed{\frac{dc_0}{dR} = -\frac{3.0457}{1.05^2} \approx -2.7625.}$$

An increase in the interest rate reduces current consumption.

9 Conclusion: Two Channels of the Income Effect

An interest-rate change affects consumption through two distinct income channels.

First, the endowment-wealth effect. A higher interest rate reduces the present value of future endowment income:

$$W = y_0 + \frac{y_1}{R}, \qquad \frac{dW}{dR} = -\frac{y_1}{R^2} < 0.$$

This reduction in wealth lowers current consumption if current consumption is normal.

Second, the demand-side income effect. Even at fixed present-value wealth, a change in the relative price of future consumption affects purchasing power. The ordinary Slutsky equation separates this demand-side income effect from the substitution effect.

For log-linear preferences, the demand-side income effect and substitution effect *exactly cancel for current consumption*. Indeed,

$$c_0(W, R) = \frac{W}{1+\beta}, \qquad \left.\frac{\partial c_0}{\partial R}\right|_W = 0.$$

Thus, only the change in endowment wealth affects current consumption:

$$\frac{dc_0}{dR} = \frac{\partial c_0}{\partial W}\frac{dW}{dR}, \qquad \boxed{\frac{dc_0}{dR} = -\frac{y_1}{(1+\beta)R^2} < 0.}$$

For our numerical example,

$$\boxed{\frac{dc_0}{dR} = -\frac{6}{1.97(1.05)^2} \approx -2.7625.}$$

Central conclusion. For log-linear preferences, substitution and demand-side income effects cancel exactly for current consumption. Its entire response to the interest rate is explained by the change in the present value of the endowment income stream.